Integracao compativel no repositorio original

Correcoes de contratos, preservacao de dados e preparacao para revisao do Tommaso.

Estado: codigo e documentacao preparados localmente. Envio direto bloqueado por permissao GitHub. Nenhum deploy ou escrita no banco oficial nesta etapa.

Base: TommasoNagliatti/EMPS-Ecosystem
`3fc3bc58fb27b48c21fc69a2ce44049365ddf7b1`

Branch:
`codex/compatibilidade-site-gie-2026-10-13`

372 testes aprovados / 8 MySQL opt-in desativados

Schema, SQL, modelos e core Python original do GIE preservados. A nova politica de limite/retirada e opt-in por instalacao; regras existentes nao sao reescritas.

Este documento nao certifica ausencia absoluta de conflitos, homologacao de equipamento ou prontidao de producao. A revisao e o checklist de release continuam obrigatorios.

Pedido: corrigir conflitos com o ecossistema original e enviar as melhorias ao GitHub do Tommaso. Publicacao proposta para revisao, sem merge/deploy automatico.

Proveniencia e preservacao

- Base original: `3fc3bc58fb27b48c21fc69a2ce44049365ddf7b1`.

- Fonte das melhorias anteriores: `2419b46b0381ffec1fee9f5848f78c5c22a3eebd` do repositorio matheus00M7/ecossistema-emps-fiap-test.

- Branch local: `codex/compatibilidade-site-gie-2026-10-13`.

- Trabalho em clone separado; instancia anterior e arquivos locais preservados.

- Historico Git original mantido, sem force push ou merge de historias desconexas.

- LICENSE e WORK_STATE original preservados; CI original ajustado, nao substituido.

- Melhorias anteriores de interface, navegacao, paineis, clima, mapas e perfis remotos transportadas com suas CRs historicas. Os PDFs anteriores descrevem seus cortes antigos, nao o estado de deploy desta nova branch.

O manifesto `docs/compatibility-manifest-2026-10-13.json` enumera o conjunto completo alterado e seus hashes. O diff Git contra a base e a fonte integral de verdade para cada linha adicionada/removida; os commits anteriores e seus relatorios permanecem referenciados, nao sao reescritos como feitos hoje.

Correcoes desta rodada

R01 - Pagamento concluido nao elimina a retirada pendente

- `app/emps-charge/src/domain/models.ts`: tipos opcionais de lifecycle e versao da politica; preco de permanencia aceita null quando indefinido.

- Novo `src/domain/charge-lifecycle.ts`: valida metadata e calcula se a retirada ainda e necessaria, independente de status financeiro ou tariffVersion.

- `src/domain/normalizers.ts`: conserva lifecycle/disconnectedAt, rejeita metadata invalida em vez de transformar silenciosamente em recarga legada.

- `src/services/mobile-api.ts`: START aceita versao opcional anunciada pela API.

- `src/context/app-context.tsx`: retirada separada de pagamento, refresh com verificacao de sessao e termos V2 enviados somente quando anunciados.

- `src/app/receipt.tsx`: acao de retirada mesmo com recibo pago, prazo e polling ate desconexao; modo fisico aguarda gateway, sem botao de retirada manual.

- `frontend/components/charge/WebCharge.tsx`: mesmas regras no browser, impede iniciar outra recarga com conector ocupado e aceita confirmacao final Stripe quando o fechamento requer acao adicional. Polling encerra ao retirar.

- `frontend/domain/charge-lifecycle.ts` e `ChargeLifecyclePanel.tsx`: modo de confirmacao fisico/cliente, sem ampliar a ponte somente leitura.

- `backend/src/session-billing.service.ts`: o endpoint de desconexao existente tambem aceita V2 pago e V2 sem ledger V1, preservando ownership, transacao, idempotencia e bloqueio de sessao fisica. Nao cria novo recebimento na retirada.

- `backend/src/charge-lifecycle.service.ts`: retirada fisica segue protegida mesmo se a URL do gateway desaparecer depois do inicio confirmado.

R02 - Leitores novos convivem com registros legados

- `backend/src/charge-lifecycle.ts`: opt-in por IDs e verificacao de termos no START; apresentacao reconhece sessoes ja persistidas com a lista desativada.

- `backend/src/main.ts`: valida configuracao de IDs no boot, sem ativacao global.

- `backend/src/tariff-engine.ts`: ledger V1 nao e inferido de JSON lifecycle-only; precificacao antiga permanece separada da nova regra.

- `backend/src/gie.service.ts`: a excecao para amostragem reconhece V2 de Caixa sem tariffVersion, mas nao altera a protecao de sessoes antigas do cutover V1.

- Documentacao explica que voltar ao backend original sem leitores compativeis nao e seguro depois de criar V2; nao promete rollback so com git checkout.

R03 - Contrato original de gateway e clientes por padrao

- `backend/src/mobile.service.ts` e `admin-operations.service.ts`: snapshot, limite local de energia, verificacao de capacidade e novo estado PREPARING sao aplicados a novas sessoes somente nas instalacoes explicitamente opt-in.

- `backend/src/mobile.dtos.ts`: politica opcional e validada. Cliente antigo permanece aceito no fluxo original; nao inicia V2 com termos desatualizados.

- `.env.example`: lista V2 vazia por padrao, com instrucao de homologacao.

- `charging-gateway.service.ts`, ja presente na fonte importada, conserva /commands para START sem limite e exige capacidade de hard limit somente V2. Nao foi implementado o gateway fisico externo nesta entrega.

R04 - Inicializacao nao reutiliza build antigo

- `scripts/start-local.ps1`: recompila Nest antes de iniciar servicos e aborta se falhar; recupera --v2 para ownership/supervisao de processos originais.

- `scripts/check-component.cjs`: verificacao da API nao herda o opt-in V2, conexao oficial ou flags de testes com escrita do processo chamador.

- Nao adicionados seed, migrate, reset, db push ou apagamento de volumes.

R05 - Condicoes exibidas correspondem a regra aplicada

- `tariff-engine.ts`: original conserva 15 minutos/R$ 0,25/teto; V2 informa 10 minutos e preco/teto de permanencia null, pois a tarifa esta pendente.

- `app/src/app/checkout.tsx`, `receipt.tsx` e WebCharge: prazos e pagamento separados da retirada, sem prometer cobranca de permanencia ainda nao definida.

Dependencias, verificacao e registro

- `frontend/package.json` e lockfile: Next de 16.3.2 para 16.3.8 exato, mantendo a serie 16.3; nao realizada migracao major de Tailwind ou Recharts.

- `app/package.json`: suite passa a incluir automaticamente tests/*.test.ts.

- `.github/workflows/ci.yml`: permissao somente contents:read e frontend roda toda a suite; nenhum comando de deploy foi acrescentado ao CI original.

- Novos testes de rollout/backend, WebCharge e normalizacao/app, sem banco oficial ou gateway real. README/indice atualizados e guia de release novo.

- Removidos espacos/finais de linha sobrando no modelo de aprovacao e nos testes account-theme/station-monitoring. O OFL.txt de terceiros foi mantido exatamente como recebido, inclusive seu espaco final original.

Banco de dados e GIE

Schema Prisma, migrations, SQL canonicamente existentes, seed, arquivos de deploy Docker e core Python do GIE comparados com a base original sem diferencas. Arquivos de politica/documentacao de database foram importados, nao executados. Nao houve DROP, TRUNCATE, migration, alteracao de grants ou importacao de registros.

O backend ganhou/adaptou regras em campos e tabelas existentes, inclusive JSON billingSnapshot e handlers de sessao. Isso e mudanca de backend, nao apenas CSS. Quando ativado num ambiente com escrita, operacoes autorizadas criam/atualizam registros normalmente; esta etapa nao realizou essas operacoes no banco oficial.

Verificacao realizada

Suite	Resultado
Backend generate, build e testes	126 aprovados, 8 testes MySQL opt-in desativados, zero falhas
Frontend Next 16.3.8, TypeScript e build	Aprovados com Turbopack e dependencias locais
Frontend testes	196 aprovados, zero falhas
App lint, TypeScript e testes	34 aprovados, zero falhas
GIE HTTP	11 aprovados, zero falhas
Estrutura/publicacao	5 aprovados; scanner sem sinal bloqueante

Total: 372 testes aprovados. Os novos casos verificam retirada apos pagamento, ownership, gateway somente no escopo V2, cutover, idempotencia e termos do START. O app nao foi testado em um iPhone real. Gateway/Stripe nos testes sao falsos. Os oito casos MySQL precisam de homologacao posterior em banco descartavel.

Next 16.3.2 tinha alerta critico; a versao fixada elimina esse alerta no audit executado, mas o frontend ainda possui 12 alertas (9 altos, 3 moderados). Nao foi usado npm audit fix --force para impor migracoes incompativeis. Isso nao e certificado de seguranca nem liberacao de producao. Referencia primaria: https://github.com/vercel/next.js/security/advisories/GHSA-p293-qw3h-jr36

Publicacao e aprovacao

Preparado para commit/branch/PR de revisao, nao para sobrescrever main. O GitHub informou falta de permissao de escrita de matheus00M7 no repositorio do Tommaso (403 no dry-run). Se isso nao mudar, o envio direto nao podera ocorrer. Conceder colaboracao e aceitar convite, ou aprovar o fluxo normal de fork/PR.

Nenhuma alteracao da Hostinger foi autorizada ou realizada nesta rodada. Merge e deploy precisam de aprovacao separada, verificacao de hooks, backup e homologacao conforme `docs/COMPATIBILITY_RELEASE.md`.

Inventario integral de arquivos

234 arquivos de codigo/documentacao diferentes da base, incluindo o PDF publico desta rodada. O manifesto JSON exclui apenas a si proprio. A/M/D significam adicionado/modificado/removido.

Hashes de cada blob Git e SHA256 constam no manifesto JSON. O manifesto exclui a si proprio para evitar um hash recursivo. O PDF publico desta rodada tambem sera anexado a branch.

Backend (37)

M | .env.example

A | .env.remote.example

M | README.md

A | scripts/remote-development.cjs

M | src/admin-operations.service.ts

M | src/app.module.ts

A | src/charge-lifecycle.controller.ts

A | src/charge-lifecycle.service.ts

A | src/charge-lifecycle.ts

M | src/charging-gateway.service.ts

A | src/dashboard-summary.ts

M | src/dashboard.controller.ts

M | src/gie.service.ts

A | src/local-gie-demo.service.ts

M | src/main.ts

M | src/mobile.dtos.ts

M | src/mobile.service.ts

M | src/notification.gateway.ts

M | src/operations.controller.ts

M | src/payment-gateway.service.ts

M | src/persistence.ts

M | src/platform-stations.service.ts

M | src/prepaid-budget.service.ts

M | src/prepaid-domain.ts

M | src/prisma.service.ts

A | src/remote-development.ts

M | src/reservations.service.ts

M | src/session-billing.service.ts

M | src/tariff-engine.ts

M | src/web-auth.service.ts

A | tests/cash-prepaid-stop.test.ts

A | tests/charge-lifecycle.test.ts

A | tests/compatibility-rollout.test.ts

A | tests/dashboard-panel.test.ts

A | tests/local-gie-demo.test.ts

A | tests/remote-development.test.ts

A | tests/station-list.test.ts

Site (121)

M | README.md

A | app/account/page.tsx

A | app/api/official-test/[...path]/route.ts

M | app/dashboard/page.tsx

M | app/layout.tsx

M | app/login/page.tsx

M | app/register/page.tsx

M | app/settings/page.tsx

A | app/stations/[id]/dashboard/page.tsx

M | app/stations/[id]/page.tsx

M | app/stations/new/page.tsx

M | app/stations/page.tsx

A | components/account/AccountSettings.tsx

A | components/account/ProfileMenu.tsx

A | components/account/account-model.ts

A | components/account/account-settings.module.css

A | components/account/profile-menu.module.css

A | components/account/use-profile-summary.ts

M | components/charge/WebCharge.tsx

M | components/chargers/CanonicalChargers.tsx

A | components/chargers/ChargeLifecyclePanel.tsx

M | components/chargers/ChargerProvisioningPage.tsx

M | components/chargers/ChargerVisualBoard.tsx

A | components/chargers/charger-list-model.ts

M | components/dashboard/AdminDashboard.tsx

A | components/dashboard/DateControl.tsx

A | components/dashboard/EnergyDiagram.tsx

A | components/dashboard/MonitoringChart.tsx

M | components/dashboard/RoleDashboard.tsx

A | components/dashboard/StationDashboard.tsx

A | components/dashboard/StationFinancialSummary.tsx

A | components/dashboard/StationNotices.tsx

A | components/dashboard/WeatherPopover.tsx

A | components/dashboard/station-dashboard.module.css

A | components/dashboard/use-station-history.ts

A | components/dashboard/use-station-weather.ts

A | components/dashboard/weather-popover.module.css

M | components/energy/EnergyFlowStation.tsx

M | components/energy/GieControls.tsx

M | components/operations/CreateAdminForm.tsx

M | components/operations/OperationalPage.tsx

A | components/operations/OperationsUI.tsx

M | components/operations/operation-config.tsx

A | components/operations/operational-model.ts

A | components/operations/operations.module.css

M | components/realtime/RealtimeProvider.tsx

M | components/shell/AppShell.tsx

A | components/shell/NavigationLink.tsx

A | components/shell/SidebarNavigation.tsx

M | components/shell/shell-navigation.ts

A | components/stations/StationEquipmentFields.tsx

A | components/stations/StationFinishFields.tsx

M | components/stations/StationList.tsx

A | components/stations/StationListControls.tsx

A | components/stations/StationListPopover.tsx

A | components/stations/StationListTable.tsx

A | components/stations/StationLocationFields.tsx

A | components/stations/StationLocationMap.tsx

M | components/stations/StationWizard.tsx

A | components/stations/StationWizardField.tsx

A | components/stations/station-geolocation.ts

A | components/stations/station-list-copy.ts

A | components/stations/station-list-model.ts

A | components/stations/station-list.module.css

A | components/stations/station-location-map-browser.ts

A | components/stations/station-location-map.module.css

A | components/stations/station-location-map.ts

A | components/stations/station-wizard-font.ts

A | components/stations/station-wizard-model.ts

A | components/stations/station-wizard-save.ts

A | components/stations/station-wizard.module.css

A | components/theme/ThemeProvider.tsx

A | components/theme/ThemeSelector.tsx

A | components/theme/theme-selector.module.css

A | components/theme/theme-store.ts

A | components/ui/Popover.tsx

A | components/ui/popover.module.css

A | domain/charge-lifecycle.ts

M | domain/emps.ts

M | domain/gie.ts

A | domain/station-insights.ts

A | domain/station-monitoring.ts

A | domain/station-weather.ts

M | next.config.ts

M | package-lock.json

M | package.json

A | public/fonts/poppins/OFL.txt

A | public/fonts/poppins/Poppins-Bold.ttf

A | public/fonts/poppins/Poppins-Medium.ttf

A | public/fonts/poppins/Poppins-Regular.ttf

A | public/fonts/poppins/Poppins-SemiBold.ttf

A | server/official-test-bridge.ts

M | services/emps-api.ts

M | services/emps-mappers.ts

A | services/navigation-cache.ts

M | styles/animations.css

M | styles/charger-board.css

M | styles/login.css

M | styles/provisioning.css

M | styles/responsive.css

M | styles/shell.css

A | styles/theme.css

A | tests/account-theme.test.cjs

A | tests/charge-lifecycle.test.cjs

A | tests/dashboard-performance.test.cjs

A | tests/date-control.test.cjs

M | tests/energy-flow.test.cjs

A | tests/local-gie-demo.test.cjs

A | tests/navigation-performance.test.cjs

A | tests/official-test-bridge.test.cjs

A | tests/operational-panels.test.cjs

A | tests/station-geolocation.test.cjs

A | tests/station-history.test.cjs

A | tests/station-insights.test.cjs

A | tests/station-list.test.cjs

A | tests/station-monitoring.test.cjs

A | tests/station-navigation.test.cjs

A | tests/station-weather.test.cjs

A | tests/station-wizard.test.cjs

A | tests/web-charge-compatibility.test.cjs

M | tsconfig.json

Aplicativo (10)

M | emps-charge/README.md

M | emps-charge/package.json

M | emps-charge/src/app/checkout.tsx

M | emps-charge/src/app/receipt.tsx

M | emps-charge/src/context/app-context.tsx

A | emps-charge/src/domain/charge-lifecycle.ts

M | emps-charge/src/domain/models.ts

M | emps-charge/src/domain/normalizers.ts

M | emps-charge/src/services/mobile-api.ts

A | emps-charge/tests/charge-lifecycle.test.ts

GIE (2)

A | GIE/service/isolated_preview.py

A | GIE/tests/service/test_isolated_preview.py

Documentacao (41)

A | CHANGE_APPROVAL_TEMPLATE.md

A | COMPATIBILITY_RELEASE.md

A | DATABASE_CHANGE_POLICY.md

M | EMPS_V2_ARCHITECTURE.md

A | README.md

A | REMOTE_DEVELOPMENT_HOSTINGER.md

M | STRIPE_LOCAL_SETUP.md

A | change-requests/2026-10-09-dashboard-cards.md

A | change-requests/2026-10-09-dashboard-sems.md

A | change-requests/2026-10-09-overview-date-controls.md

A | change-requests/2026-10-09-settings-consolidation.md

A | change-requests/2026-10-09-weather-popups.md

A | change-requests/2026-10-10-monitoring-charts.md

A | change-requests/CR-2026-10-07-001.md

A | change-requests/CR-2026-10-07-002.md

A | change-requests/CR-2026-10-07-003.md

A | change-requests/CR-2026-10-08-001.md

A | change-requests/CR-2026-10-08-002.md

A | change-requests/CR-2026-10-08-003.md

A | change-requests/CR-2026-10-08-004.md

A | change-requests/CR-2026-10-08-005.md

A | change-requests/CR-2026-10-08-006.md

A | change-requests/CR-2026-10-10-002.md

A | change-requests/CR-2026-10-10-004-performance.md

A | change-requests/CR-2026-10-11-002-isolated-gie.md

A | change-requests/CR-2026-10-12-001-operational-panels.md

A | change-requests/CR-2026-10-12-002-github-publication.md

A | change-requests/CR-2026-10-12-003-navigation-performance.md

A | change-requests/CR-2026-10-12-004-dashboard-insights.md

A | change-requests/CR-2026-10-12-006-disable-isolated-gie.md

A | change-requests/CR-2026-10-13-001-readonly-cashier.md

A | change-requests/CR-2026-10-13-002-official-cash-test.md

A | change-requests/CR-2026-10-13-003-prepaid-cash-stop.md

A | change-requests/CR-2026-10-13-004-budget-refund-removal.md

A | change-requests/CR-2026-10-13-005-github-publication.md

A | change-requests/CR-2026-10-13-006-compatibility-integration.md

A | change-requests/PROPOSTA-2026-10-12-005-integracao-oficial.md

A | change-requests/pdf/EMPS_INTEGRACAO_COMPATIVEL_TOMMASO_2026-10-13.pdf

A | change-requests/pdf/EMPS_ORGANIZACAO_GITHUB_2026-10-12.pdf

A | change-requests/pdf/EMPS_RELATORIO_COMPLEMENTAR_PUBLICO_2026-10-13.pdf

A | publication-manifest-2026-10-12.json

Raiz, scripts e database (23)

A | .gitattributes

M | .github/workflows/ci.yml

M | .gitignore

M | README.md

M | STARTUP.md

M | START_EMPS_V2.ps1

M | STOP_EMPS_V2.ps1

A | package.json

A | scripts/check-component.cjs

A | scripts/check-publication.cjs

A | scripts/check-remote-dev.ps1

M | scripts/emps-v2-processes.ps1

M | scripts/start-local.ps1

A | scripts/start-official-cash-test.ps1

A | scripts/start-remote-dev.ps1

A | scripts/stop-remote-dev.ps1

A | scripts/tests/publication.test.cjs

A | scripts/tests/test-remote-stop.ps1

M | site/emps-site-main/.gitignore

M | site/emps-site-main/README.md

M | site/emps-site-main/STARTUP.md

M | site/emps-site-main/database/README.md

A | site/emps-site-main/database/patches/README.md